AGRIM JOSHI

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EDUCATION

Simon Fraser University — Burnaby, BC **Expected 12/2027**

Bachelor of Applied Science, Computing Science

PROFESSIONAL EXPERIENCE

Special Projects Coordinator – Software Developer **Jan 2026 – Present**

SFU Beedie School of Business — Burnaby, BC *Co-op: Jan–Apr 2026 | Contractor (Part-Time): Sept 2026- Present*

- Developed a **full-stack degree planning web application** that helps SFU Beedie students choose courses, check degree requirements, and plan future semesters.
- Built the **frontend and backend using PHP** and designed a **PostgreSQL database** to manage 181 courses, degree requirements, concentrations, prerequisites, and student selections.
- Developed **REST APIs** to connect the web application with other systems and support application integrations, and created tools to summarize expected course demand, helping administrators improve semester planning.

Programming Instructor **Sept 2025 – Dec 2025**

Wize Computing Academy — Vancouver, BC

- Designed and delivered interactive lessons on Python fundamentals, covering algorithms, problem-solving, and debugging.
- Introduced 3D modelling tools to build creativity and spatial reasoning, preparing students for engineering and design applications.

TECHNICAL PROJECTS

ShadowAthlete - Personalized AI-powered Sports Analytics and Training Application **Oct 2026**

Python, FastAPI, React Native, PostgreSQL, MMpose, YOLOv8, OpenCV, PyTorch, REST APIs, JWT, Postman, Docker

- Built a **full-stack AI** sports training platform that analyzes athlete videos across 4 different sports and generates an evolving digital avatar which evolves as the user improves.
- Developed a computer vision pipeline using **MMpose, YOLOv8 and PyTorch** that extracts 33 body landmarks per frame and converts that data into multiple performance metrics, processing a 30-second clip in 10 seconds.
- Engineered a **FastAPI backend with 15+ REST modules**, **PostgreSQL**, **JWT authentication**, and **Postman API testing**, containerized with **Docker** and integrated with the **React Native** frontend.

Re-Habit — AI Motion-Tracking Web-App for Physiotherapy **Oct 2025**

Python, Flask, MediaPipe, OpenCV, Pose Estimation

- Built a **camera-based web application** that tracks a patient's body movements in real time and guides them through stroke-recovery exercises.
- Used **MediaPipe** and **OpenCV** to track joints, count exercise repetitions, and detect incorrect movements while reducing false counts.
- Created a dashboard that measures movement symmetry, posture, and walking patterns and turns the results into simple progress summaries.

Financial Advisor Application - Web App with real-time Financial Insights **Jan 2025**

React, Node.js, Docker, REST APIs, CI/CD, Docker

- Built a **full-stack financial advisor web application** using **React and Node.js**, integrating real-time financial APIs to retrieve account data and generate personalized financial insights.
- Engineered **REST API integrations and backend workflows** with automated API validation and regression testing, optimized backend performance through API request caching and reduced redundant network calls, improving response efficiency.
- Designed and deployed an automated **CI/CD pipeline using GitHub Actions and Docker** to build, test and deploy application changes, reducing the release cycle by **60%**.

SKILLS

Languages/Database	: C/C++, Rust, Golang, Python, JavaScript, HTML, CSS, SQL, PHP, PostgreSQL, MongoDB, R, MATLAB
Frameworks/ Libraries	: React, React Native, Node.js, Flask, Pandas, NumPy
AI/ML	: TensorFlow, PyTorch, OpenCV, Pose Estimation, Scikit-Learn, Data Annotation, YOLO, Mediapipe
Web/API & Security	: REST APIs, FastAPI, OAuth 2.0, JWT, CORS, WebSockets, JSON, API Authentication, TCP/IP, UDP
Tools	: Git, GitHub, Docker, Postman, Redis, Linux, Kafka, CLI, Neovim, Figma

ACHIEVEMENTS & CERTIFICATES

- Placed Top 3 at the Stormhacks hackathon, for the project Re-Habit, out of 250+ teams and 1000+ students.
- Data Analyst Certificate, Coursera (Oct 2024) | Data Structures & Algorithms, ITI Informatics (Aug–Dec 2023).